



2012 Yearly Examination

FORM V

MATHEMATICS EXTENSION 1

Wednesday 29th August 2012

General Instructions

- Writing time — 3 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 117 Marks

- All questions may be attempted.

Section I — 13 Marks

- Questions 1–13 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II — 104 Marks

- Questions 14–21 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your name, class and master on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Fourteen.
- Write your name and master on this question paper and submit it with your answers.

5A: DS

5D: DNW

5G: SO

5B: TCW

5E: LYL

5H: BR

5C: REP

5F: MLS

5I: SJE

Checklist

- SGS booklets — 8 per boy
- Multiple choice answer sheet
- Candidature — 1000 boys

Examiner

DNW

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

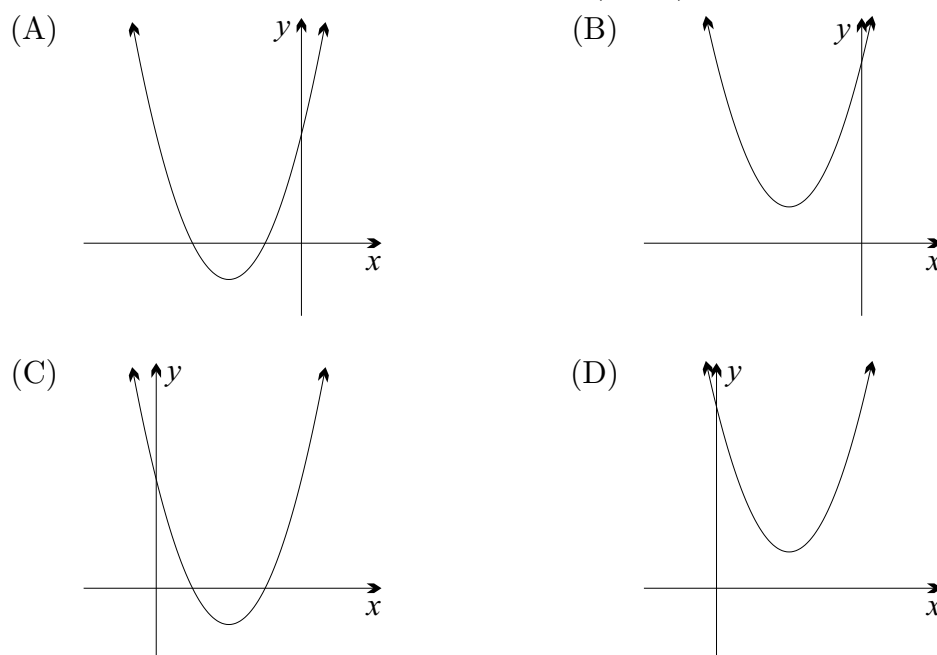
QUESTION ONE

The factors of $3x^2 - 10x - 8$ are:

- (A) $(3x + 4)(x - 2)$ (B) $(3x - 4)(x + 2)$
 (C) $(3x - 2)(x + 4)$ (D) $(3x + 2)(x - 4)$

QUESTION TWO

Which of the following graphs best represents $y = (x + 2)^2 - 1$?



QUESTION THREE

The derivative of $3x^4 + x^5$ is:

- (A) $\frac{3}{5}x^5 + \frac{1}{6}x^6$ (B) $3x^3 + x^4$ (C) $12x^3 + 5x^4$ (D) $12x^3 + x^5$

QUESTION FOUR

The derivative of e^{3x} is:

- (A) $\frac{1}{3}e^{3x}$ (B) e^{3x} (C) $3xe^{3x-1}$ (D) $3e^{3x}$

QUESTION FIVE

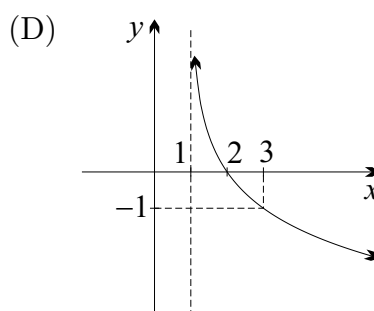
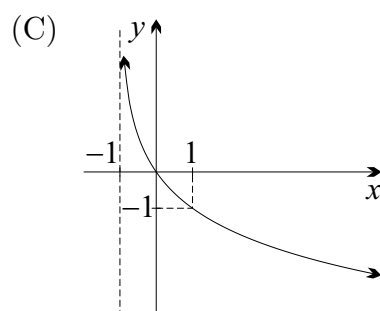
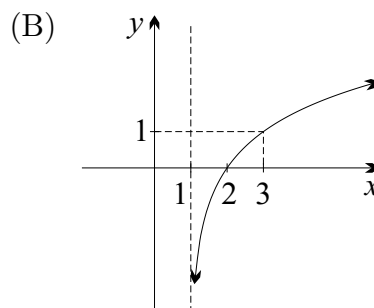
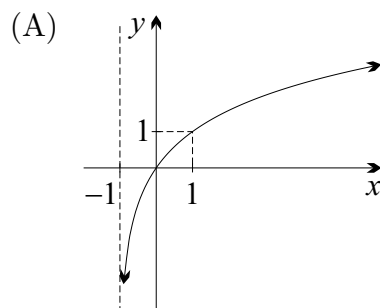
When $3 \log p + \log(2q)$ is simplified, the result is:

- (A) $\log(6pq)$ (B) $\log(2p^3q)$ (C) $\log(3pq^2)$ (D) $\log(p^3q^2)$

Exam continues next page ...

QUESTION SIX

Which of the following is the graph of $y = -\log_2(x + 1)$?



QUESTION SEVEN

Here is a table of values for $y = 2^{-x^2}$.

x	0	1	2
y	1	$\frac{1}{2}$	$\frac{1}{16}$

Applying Simpson's rule to these values, an estimate of $\int_0^2 2^{-x^2} dx$ is:

- (A) $\frac{49}{96}$ (B) $\frac{49}{48}$ (C) $\frac{33}{32}$ (D) $\frac{49}{24}$

QUESTION EIGHT

The indefinite integral $\int (2x + 1)^3 dx$ is equal to:

- (A) $(2x + 1)^4 + C$ (B) $\frac{(2x + 1)^4}{2} + C$
 (C) $\frac{(2x + 1)^4}{4} + C$ (D) $\frac{(2x + 1)^4}{8} + C$

QUESTION NINE

The quadratic $Q(x) = ax^2 + bx + c$ is negative definite. Which of the following is true?

- (A) $a > 0$ and $\Delta < 0$ (B) $a < 0$ and $\Delta < 0$
 (C) $a > 0$ and $\Delta > 0$ (D) $a < 0$ and $\Delta > 0$

Exam continues overleaf ...

QUESTION TEN

The derivative of $\sqrt{3x^2 - 1}$ is:

- (A) $\frac{x}{\sqrt{3x^2 - 1}}$ (B) $\frac{2x}{\sqrt{3x^2 - 1}}$ (C) $\frac{3x}{\sqrt{3x^2 - 1}}$ (D) $\frac{6x}{\sqrt{3x^2 - 1}}$

QUESTION ELEVEN

It is known that $f''(x) = (x - 1)^2(x + 1)$.

How many inflexion points does the graph of $y = f(x)$ have?

- (A) 0 (B) 1 (C) 2 (D) 3

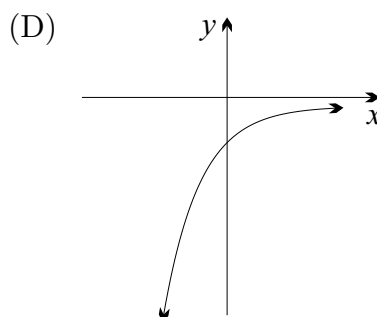
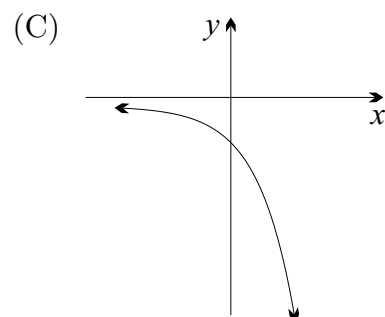
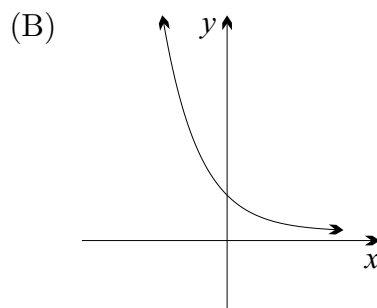
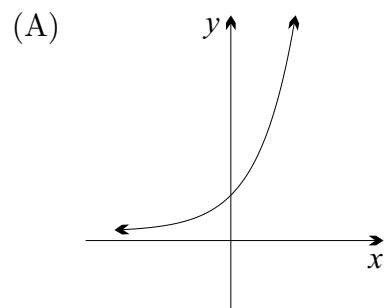
QUESTION TWELVE

What is the value of $\int_1^2 e^{x-1} dx$?

- (A) $\frac{1}{2}e^2 - e$ (B) $e - 1$ (C) e (D) 1

QUESTION THIRTEEN

Suppose that $f'(x) > 0$ and $f''(x) < 0$ for all real values of x . Which of the following graphs best represents $y = f(x)$?



_____ End of Section I _____

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION FOURTEEN (13 marks) Use a separate writing booklet. **Marks**

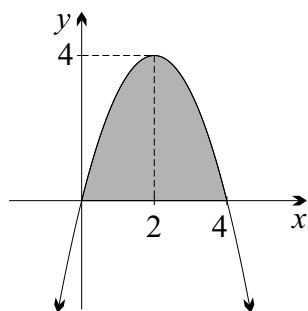
- | | |
|---|---|
| (a) Simplify $ -3 - 7 $. | 1 |
| (b) Determine the exact value of $\cos 150^\circ$. | 1 |
| (c) Evaluate $\log_2 8$. | 1 |
| (d) Solve $3 - 2x \geq 7$. | 2 |
| (e) Find a and b if $(5 - \sqrt{2})^2 = a + b\sqrt{2}$. | 2 |
| (f) Write down the primitive of $x^2 + 3$. | 2 |
| (g) Express $\frac{1}{3 - \sqrt{2}} + \frac{1}{3 + \sqrt{2}}$ in simplest form. | 2 |
| (h) Determine the coordinates of the mid-point of AB , where $A = (3, -4)$ and $B = (7, 2)$. | 2 |

QUESTION FIFTEEN (13 marks) Use a separate writing booklet. **Marks**

- | | |
|---------------------------------|---|
| (a) Differentiate: | |
| (i) $(3x^2 + 4)^5$ | 2 |
| (ii) $x \log x$ | 2 |
| (iii) $\frac{x}{3x + 1}$ | 2 |
| (b) Evaluate: | |
| (i) $\int_1^2 \frac{1}{x^2} dx$ | 2 |
| (ii) $\int_0^1 e^{2x+1} dx$ | 2 |

Exam continues overleaf ...

(c)



3

The graph above shows the shaded region bounded by the x -axis and the parabola $y = 4x - x^2$. Find the area of this region.

QUESTION SIXTEEN (13 marks) Use a separate writing booklet.

Marks

(a) Determine the gradient of the tangent to $y = 2x^2 - x^3$ at the point where $x = 2$.

2

(b) Find the coordinates of the vertex of the parabola with equation $y = (x - 4)(x + 1)$.

2

(c) Let $x = \log_a 5$ and $y = \log_a 3$. Write $\log_a 45$ in terms of x and y .

2

(d) Consider the integral $I = \int_1^2 \ln x \, dx$.

(i) Find the approximate value of I using the trapezoidal rule with three function values. Give your answer correct to 2 decimal places.

3

(ii) Give a reason why the answer to part (i) is less than the exact value of I .

1

(e) Show that $\int_0^4 \frac{x}{x^2 + 9} \, dx = \log \frac{5}{3}$.

3

QUESTION SEVENTEEN (13 marks) Use a separate writing booklet.

Marks

(a) Solve $3 \tan^2 \theta - 5 \sec \theta + 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$.

4

Approximate your answers to the nearest degree where necessary.

(b) The region between $y = \frac{1}{\sqrt{x}}$ and the x -axis, for $1 \leq x \leq e^2$, is rotated about the x -axis to generate a solid of revolution. Find the exact volume of this solid.

3

(c) (i) Differentiate $y = xe^x$.

1

(ii) Hence evaluate $\int_{-1}^1 xe^x \, dx$.

3

(d) Find $\int (3x + 1)e^{3x^2 + 2x + 1} \, dx$.

2

Exam continues next page ...

QUESTION EIGHTEEN (13 marks) Use a separate writing booklet.

Marks

- (a) Consider the function $y = x - \log(x + 1)$, where $x > -1$.
You may assume that there is a vertical asymptote at $x = -1$ with $y \rightarrow \infty$.

- (i) Find and classify any stationary points.
(ii) Explain why the curve never changes concavity.
(iii) Sketch a graph of $y = x - \log(x + 1)$.
(iv) Hence solve $\log(1 + x) \geq x$.

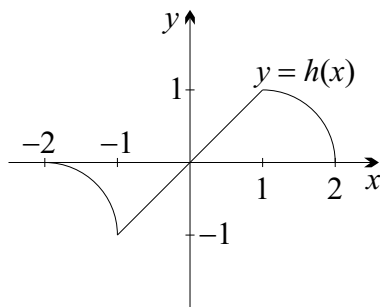
2

1

1

1

(b)



3

The graph of $y = h(x)$, shown above for $-2 \leq x \leq 2$, consists of a straight line and two quadrants. Use geometrical formulae to evaluate $\int_{-2}^2 h(x) dx$.

- (c) (i) Find the values of a , b and c if

$$x^2 + 1 \equiv a(x - 1)^2 + b(x - 1) + c$$

for all values of x .

- (ii) Hence determine $\int \frac{x^2 + 1}{(x - 1)^2} dx$.

3

2

QUESTION NINETEEN (13 marks) Use a separate writing booklet.

Marks

- (a) (i) What is the equation of a line through $(4, -4)$ with gradient m ?

1

- (ii) Suppose that the line in part (i) is tangent to $y = \frac{2}{x}$.

3

Use the discriminant to find the possible values of m .

- (b) Solve the following by first reducing it to a quadratic equation.

4

$$3 \left(x + \frac{1}{x} \right)^2 - 16 \left(x + \frac{1}{x} \right) + 20 = 0$$

- (c) By using the substitution $u = x^2 + 1$, or otherwise, determine $I = \int \frac{2x}{\sqrt{x^2 + 1}} dx$.

2

- (d) In a certain geometric sequence, the sum of the first two terms is 8 and the sum of the first three terms is 26. Find the possible values of the common ratio.

3

Exam continues overleaf ...

QUESTION TWENTY (13 marks) Use a separate writing booklet.

Marks

(a) (i) Find the equation of the tangent to $y = \log x$ at the point $(a, \log a)$.

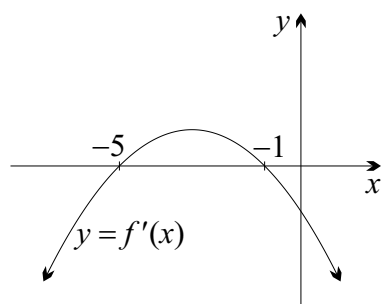
2

(ii) This tangent passes through the origin. Find the value of a .

1

(b)

2



The graph above shows the gradient function of the curve $y = f(x)$.

What is the value of x for which the graph of $y = f(x)$ has a maximum turning point. Justify your answer.

(c) Factorise $p^3 + q^3$.

1

(d) The quadratic equation $2x^2 - 3x - 4 = 0$ has roots p and q .

(i) Without solving the equation, determine:

(α) $p + q$

1

(β) pq

1

(γ) $p^3 + q^3$

1

(ii) Hence or otherwise find a quadratic equation with integer coefficients which has roots p^3 and q^3 .

1

(e) The function $f(t)$ is even and hence

3

$$\int_{-x}^x f(t) dt = 2 \int_0^x f(t) dt.$$

Let $F(x) = \int_0^x f(t) dt.$

By considering $F(x) - F(-x)$, and using the properties of definite integrals, show that $F(x)$ is odd.

QUESTION TWENTY ONE (13 marks) Use a separate writing booklet.

Marks

- (a) The function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} e^{2x} & \text{for } x < 0 \\ ax^2 + bx + c & \text{for } x \geq 0 \end{cases}$$

It is known that $f(x)$ is continuous and differentiable for all real values of x .

It is also known that $f(1) = 0$.

- | | |
|---|---|
| (i) Find $f'(x)$. | 1 |
| (ii) Show that $a = -3$, $b = 2$ and $c = 1$. | 3 |
| (iii) Sketch a graph of $y = f(x)$. | 1 |
| (iv) Hence determine the global maximum of $f(x)$. | 1 |
- (b) (i) Write a^x as a power of e .
- 1
- (ii) Hence show that $\frac{d}{dx}(a^x) = a^x \log a$.
- 1
- (c) Let $g(x) = a^x - x^a$ where $a > e$ and is constant, and $x \geq 0$.
 You may assume that $g(x)$ is continuous for all $x \geq 0$.
 Note that $g(a) = 0$.
- | | |
|---|---|
| (i) Evaluate $g(0)$. | 1 |
| (ii) Show that $g'(a) > 0$. | 1 |
| (iii) Explain why $y = g(x)$ has at least two x -intercepts. | 1 |
| (iv) In this part assume that $y = g(x)$ has exactly two x -intercepts. One is at $x = a$.
Let the other be at $x = b$. By considering the sign of $g'(b)$, show that | 2 |

$$b < \frac{a}{\log a}.$$

————— End of Section II —————

END OF EXAMINATION

B L A N K P A G E

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



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FORM V
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Wednesday 29th August 2012

- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

NAME:

CLASS: MASTER:

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

Question Eleven

A ☐ B ☐ C ☐ D ☐

Question Twelve

A ☐ B ☐ C ☐ D ☐

Question Thirteen

A ☐ B ☐ C ☐ D ☐

Multiple Choice (with explanations of errors)

- Q 1** (D) (A) $3x^2 - 2x - 8$, (B) $3x^2 + 2x - 8$, (C) $3x^2 + 10x - 8$
- Q 2** (A) (B) $y = (x + 2)^2 + 1$, (C) $y = (x - 2)^2 - 1$, (D) $y = (x - 2)^2 + 1$
- Q 3** (C) (A) primitive, (B) failure to multiply by index, (D) only first term
- Q 4** (D) (A) primitive, (B) f' missing from $f'e^f$, (C) confused with $\frac{d}{dx}(x^3)$
- Q 5** (B) (A) $3 \log p \neq \log(3p)$, (C) $3 \log p \neq \log(3p)$ and $\log(2q) \neq \log(q^2)$,
(D) $\log(2q) \neq \log(q^2)$
- Q 6** (C) (A) $y = \log_2(x + 1)$, (B) $y = \log_2(x - 1)$, (D) $y = -\log_2(x - 1)$
- Q 7** (B) (A) $\frac{1}{6}h(f_0 + 4f_1 + f_2)$, (C) $\frac{1}{2}h(f_0 + 2f_1 + f_2)$ (Trapezoidal rule)
(D) $\frac{2}{3}h(f_0 + 4f_1 + f_2)$
- Q 8** (D) (A) $\int 8(2x + 1)^3 dx$ (B) $\int 4(2x + 1)^3 dx$ (C) $\int 2(2x + 1)^3 dx$
- Q 9** (B) (A) positive definite (C) indefinite (D) indefinite
- Q 10** (C) (A) $\frac{d}{dx}\left(\frac{1}{3}\sqrt{3x^2 - 1}\right)$ (B) $\frac{d}{dx}\left(\frac{2}{3}\sqrt{3x^2 - 1}\right)$ (D) $\frac{d}{dx}\left(2\sqrt{3x^2 - 1}\right)$
- Q 11** (B) (A) f'' changes sign at $x = -1$ (C) f'' does not change sign at $x = 1$
(D) f'' only has two zeros
- Q 12** (B) (A) $\int e^{x-1} dx \neq \frac{e^x}{x}$ (C) $e^0 \neq 0$ (D) $\int e^{x-1} dx \neq (x - 1)e^{x-2}$
- Q 13** (D) (A) $f'' > 0$ (B) $f'' > 0, f' < 0$ (C) $f' < 0$

QUESTION FOURTEEN (13 marks)

(a) $|-3| - |7| = -4$ ☒

(b) $\cos 150^\circ = -\frac{\sqrt{3}}{2}$ ☒

(c) $\log_2 8 = 3$ ☒

(d) $3 - 2x \geq 7$
 $-2x \geq 4$ ☒
 $x \leq -2$ ☒

(e) $(5 - \sqrt{2})^2 = 25 - 10\sqrt{2} + 2$ ☒
 $= 27 - 10\sqrt{2}$ ☒
 so $a = 27$ and $b = -10$.

(f) $\int x^2 + 3 dx = \frac{1}{3}x^3 + 3x + C$ ☒
 [2nd mark for constant.]

(g) $\frac{1}{3 - \sqrt{2}} - \frac{1}{3 + \sqrt{2}} = \frac{3 + \sqrt{2} - (3 - \sqrt{2})}{3^2 - 2}$ ☒
 $= \frac{2\sqrt{2}}{7}$ ☒

(h) mid-point $= \left(\frac{3+7}{2}, \frac{-4+2}{2} \right)$ ☒
 $= (5, -1)$ ☒

Total for Question 14: 13 Marks

QUESTION FIFTEEN (13 marks)

(a) (i) $\frac{d}{dx} (3x^2 + 4)^5 = 5 \times (3x^2 + 4)^4 \times 3x \times 2$ (chain rule) ☒
 $= 30x(3x^2 + 4)^4$ ☒

(ii) $\frac{d}{dx} (x \log x) = 1 \times \log x + x \times \frac{1}{x}$ (product rule) ☒
 $= \log x + 1$ ☒

(iii) $\frac{d}{dx} \left(\frac{x}{3x+1} \right) = \frac{(3x+1) \times 1 - x \times 3}{(3x+1)^2}$ (quotient rule) ☒
 $= \frac{1}{(3x+1)^2}$ ☒

(b) (i) $\int_1^2 \frac{1}{x^2} dx = \left[-\frac{1}{x} \right]_1^2$ ☒
 $= -\frac{1}{2} + 1$ ☒
 $= \frac{1}{2}$ ☒

(ii) $\int_0^1 e^{2x+1} dx = \left[\frac{1}{2} e^{2x+1} \right]_0^1$ ☒
 $= \frac{1}{2} e^3 - \frac{1}{2} e$ ☒
 $= \frac{1}{2} e(e^2 - 1).$ ☒

(c) Area $= \int_0^4 4x - x^2 dx$ ☒
 $= \left[2x^2 - \frac{1}{3}x^3 \right]_0^4$ ☒
 $= \frac{32}{3}.$ ☒

Total for Question 15: 13 Marks

QUESTION SIXTEEN (13 marks)

(a) $y = 2x^2 - x^3$
 so $\frac{dy}{dx} = 4x - 3x^2$ ☒
 at $x = 2$: $\frac{dy}{dx} = 4 \times 2 - 3 \times 2^2$ ☒
 $= -4.$ ☒

(b) x -intercepts $= -1, 4$ ☒
 so vertex is at $x = \frac{-1+4}{2} = \frac{3}{2}$
 where $y = -6\frac{1}{4}$ ☒
 [or any other valid method.]

(c) $\log_a 45 = \log_a 3^2 + \log_a 5$ ☒
 $= 2 \log_a 3 + \log_a 5$
 $= 2y + x$ ☒

(d) (i) Let $y = \ln x$ then:

x	1	$\frac{3}{2}$	2
y	0	$\ln \frac{3}{2}$	$\ln 2$

☒

so $I \doteq \frac{(\frac{1}{2})}{2} \left(0 + 2 \times \ln \frac{3}{2} + \ln 2 \right)$ ☒
 $\doteq 0.38$ (to two decimal places) ☒

(ii) $y' = \frac{1}{x}$ so $y'' = -\frac{1}{x^2}$. Thus $y'' < 0$ for all x in the domain and

the curve is concave down.



$$\begin{aligned}
 \text{(e)} \quad \int_0^4 \frac{x}{x^2+9} dx &= \frac{1}{2} \left[\log(x^2+9) \right]_0^4 \\
 &= \frac{1}{2}(\log 25 - \log 9) \\
 &= \log 5 - \log 3 \\
 &= \log \frac{5}{3}.
 \end{aligned}$$



Total for Question 16: 13 Marks

QUESTION SEVENTEEN (13 marks)

(a) $3 \tan^2 \theta - 5 \sec \theta + 1 = 0$

so $3(\sec^2 \theta - 1) - 5 \sec \theta + 1 = 0$

$$3 \sec^2 \theta - 5 \sec \theta - 2 = 0$$

$$(3 \sec \theta + 1)(\sec \theta - 2) = 0$$

thus $\sec \theta = -\frac{1}{3}$ or 2

$\sec \theta = -\frac{1}{3}$ has no real solutions.

$\sec \theta = 2$ has solutions $\theta = 60^\circ$ or 300° .



$$\begin{aligned}
 \text{(b)} \quad \text{Volume} &= \pi \int_1^{e^2} y^2 dx \\
 &= \pi \int_1^{e^2} \frac{1}{x} dx \\
 &= \pi \left[\log x \right]_1^{e^2} \\
 &= \pi(\log e^2 - \log 1) \\
 &= 2\pi.
 \end{aligned}$$



(c) (i) $y = xe^x$

$$\frac{dy}{dx} = 1 \times e^x + x \times e^x \quad (\text{product rule})$$

$$= e^x + xe^x$$



(ii) Rearrange part (i) to get

$$\begin{aligned}
 xe^x &= \frac{dy}{dx} - e^x \\
 \text{so } \int_{-1}^1 xe^x dx &= \int_{-1}^1 \frac{dy}{dx} - e^x dx && \boxed{\checkmark} \\
 &= \left[y - e^x \right]_{-1}^1 && \boxed{\checkmark} \\
 &= (e - e) - (-e^{-1} - e^{-1}) \\
 &= 2e^{-1} && \boxed{\checkmark}
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad \int (3x+1)e^{3x^2+2x+1} dx &= \frac{1}{2} \int (6x+2)e^{3x^2+2x+1} dx && \boxed{\checkmark} \\
 &= \frac{1}{2} e^{3x^2+2x+1} + C && \boxed{\checkmark}
 \end{aligned}$$

[Do not penalise lack of a constant.]

Total for Question 17: 13 Marks

QUESTION EIGHTEEN (13 marks)

(a) (i) $y = x - \log(x+1)$

$$\begin{aligned}
 \text{so } y' &= 1 - \frac{1}{x+1} \\
 &= \frac{x}{x+1}
 \end{aligned}$$

thus there is a stationary point at $(0, 0)$.

$$y'' = \frac{1}{(x+1)^2}$$

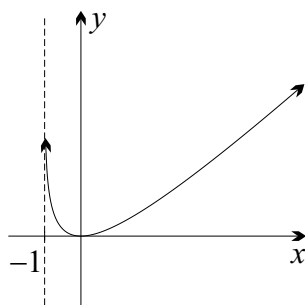
so at $x = 0$, $y'' = 1$ and it is a minimum stationary point.

(ii) $y'' > 0$ for all x in the domain,

thus y'' never changes sign,

hence the concavity never changes.

(iii)



(iv) If $\log(x+1) \geq x$

then $0 \geq x - \log(x+1)$

which, from the graph, is only true when $x = 0$.

(b) In this case, areas below the x -axis are negative,

$$\text{area quadrant} = \frac{\pi}{4}$$

$$\text{area triangle} = \frac{1}{2},$$

$$\begin{aligned} \text{hence} \quad \int_2^2 h(x) dx &= -\left(1 - \frac{\pi}{4} + \frac{1}{2}\right) + \left(\frac{1}{2} + \frac{\pi}{4}\right) \\ &= \frac{\pi}{2} - 1 \end{aligned}$$



(c) (i) Since $x^2 + 1 \equiv a(x - 1)^2 + b(x - 1) + c$

$$\text{at } x = 1: \quad 2 = c$$

$$\text{at } x = 2: \quad 5 = a + b + c$$

$$\text{at } x = 0: \quad 1 = a - b + c$$

solving simultaneously,

$$a = 1$$

$$\text{and} \quad b = 2$$



(ii) From part (i):

$$\begin{aligned} \int \frac{x^2 + 1}{(x - 1)^2} dx &= \int \frac{(x - 1)^2 + 2(x - 1) + 2}{(x - 1)^2} dx \\ &= \int 1 + \frac{2}{x - 1} + \frac{2}{(x - 1)^2} dx \\ &= x + 2 \log(x - 1) - \frac{2}{x - 1} + C \end{aligned}$$



[Do not penalise lack of a constant.]

Total for Question 18: 13 Marks

QUESTION NINETEEN (13 marks)

(a) (i) $y = mx - 4(m + 1)$

[or equivalent.]



(ii) Substitute $y = \frac{2}{x}$ into part (i) to get:

$$\frac{2}{x}mx - 4(m + 1)$$

$$\text{or} \quad mx^2 - 4(m + 1)x - 2 = 0$$

$\Delta = 0$ since the two are tangent, so:

$$16(m + 1)^2 + 8m = 0$$

$$\text{or} \quad 2m^2 + 5m + 2 = 0$$

$$\text{so} \quad (2m + 1)(m + 2) = 0$$

$$\text{thus} \quad m = -\frac{1}{2} \text{ or } -2.$$



(b) $3\left(x + \frac{1}{x}\right)^2 - 16\left(x + \frac{1}{x}\right) + 20 = 0.$

Put $\lambda = \left(x + \frac{1}{x}\right)$ to get:

$$3\lambda^2 - 16\lambda + 20 = 0$$

or $(3\lambda - 10)(\lambda - 2) = 0$ (or equivalent.) ☒

thus $\lambda = 2$ or $\frac{10}{3}.$ ☒

When $\lambda = 2$ $x + \frac{1}{x} = 2$

$$x^2 - 2x + 1 = 0$$

so $x = 1.$ ☒

When $\lambda = \frac{10}{3}$ $x + \frac{1}{x} = \frac{10}{3}$

$$3x^2 - 10x + 3 = 0$$

$$(3x - 1)(x - 3) = 0$$

so $x = 3$ or $\frac{1}{3}.$ ☒

(c) $I = \int \frac{2x}{\sqrt{x^2 + 1}} dx$
 $= 2 \int \frac{x}{\sqrt{x^2 + 1}} dx$
 $= 2\sqrt{x^2 + 1} + C$



(d) $a(r^2 + r + 1) = 26$

$$a(r + 1) = 8$$

Dividing the first by the second:

$$\frac{r^2 + r + 1}{r + 1} = \frac{13}{4}$$

so $4r^2 + 4r + 4 = 13r + 13$

or $4r^2 - 9r - 9 = 0$

thus $(4r + 3)(r - 3) = 0$

hence $r = -\frac{3}{4}$ or 3 ☒

Total for Question 19: 13 Marks

QUESTION TWENTY (13 marks)

(a) (i) $y = \log x$

so $y' = \frac{1}{x}$

and $y'(a) = \frac{1}{a}$ ☒

Thus the tangent has equation:

$y = \frac{1}{a}x + \log a - 1$ (or equivalent) ☒

(ii) At the origin:

$0 = \log a - 1$

so $\log a = 1$

or $a = e$. ☒

(b) There is a stationary point at $x = -1$ where $f'(x) = 0$. ☒Since the sign of f' changes from positive to negative, ☒

it is a maximum stationary point (local maximum).

(c) $p^3 + q^3 = (p + q)(p^2 - pq + q^2)$ ☒

(d) (i) $(\alpha) \quad p + q = \frac{-b}{a}$
 $= \frac{3}{2}$ ☒

$(\beta) \quad pq = \frac{c}{a}$
 $= -2$ ☒

$(\gamma) \quad p^3 + q^3 = (p + q)((p + q)^2 - 3pq)$
 $= \frac{99}{8}$ ☒

(ii) $(pq)^3 = -8$, so an equation with those roots is

$x^2 - \frac{99}{8}x - 8 = 0$

hence $8x^2 - 99x - 64 = 0$. ☒

$$\begin{aligned}
 (e) \quad F(x) - F(-x) &= \int_0^x f(t) dt - \int_0^{-x} f(t) dt \\
 &= \int_0^x f(t) dt + \int_{-x}^0 f(t) dt \quad (\text{reversing the direction}) \\
 &= \int_{-x}^x f(t) dt \quad (\text{combining regions}) \\
 &= 2 \int_0^x f(t) dt \\
 &= 2F(x).
 \end{aligned}$$

Hence $-F(-x) = F(x)$, ☒

that is, $F(x)$ is odd.

Total for Question 20: 13 Marks

QUESTION TWENTY ONE (13 marks)

(a) (i) $f'(x) = \begin{cases} 2e^{2x} & \text{for } x < 0 \\ 2ax + b & \text{for } x > 0 \end{cases}$ ☒

(ii) $f(x)$ is continuous so $f(0) = \lim_{x \rightarrow 0^-} f(x)$

i.e. $c = 1$ ☒

$f(x)$ is differentiable so $\lim_{x \rightarrow 0^-} f'(x) = \lim_{x \rightarrow 0^+} f'(x)$

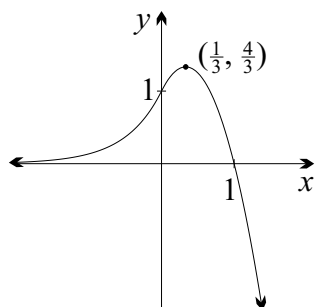
i.e. $2 = b$ ☒

$f(1) = 0$ so

$a + b + c = 0$

thus $a = -3$ ☒

(iii)



[Vertex coordinates may be omitted.] ☒

(iv) The global maximum is at the vertex of the parabola, where $x = \frac{1}{3}$

thus $f_{\max} = \frac{4}{3}$ ☒

(b) (i) $a^x = e^{x \log a}$ ☒

(ii) $y = a^x$

$= e^{x \log a}$

so $y' = e^{x \log a} \times \log a$

$= a^x \log a$ ☒

(c) For $g(x) = a^x - x^a$

(i) $g(0) = a^0 - 0^a \quad (a > 0)$

$= 1$ ☒

(ii) $g'(x) = a^x \log a - ax^{a-1}$ (by part (i))

so $g'(a) = a^a \log a - aa^{a-1}$
 $= a^a(\log a - 1).$

Now $a > e$

so $\log a > 1$

hence $g'(a) > 0$.



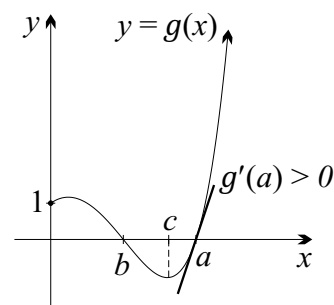
- (iii) $g(a) = 0$ so $x = a$ is an x -intercept
 $g'(a) > 0$ so there is at least one value $x = c$,
 $c < a$, for which $g(c) < 0$.

But $g(0) = 1$ and $g(x)$ is continuous.

Thus $g(x)$ changes sign between c and 0 .

Hence $g(x)$ must have another x -intercept.

[Or any other valid argument.]



- (iv) It follows that $g'(b) < 0$.

now $g'(b) = a^b \log a - ab^{a-1}$

$$= a^b \log a - \frac{a}{b} \times b^a$$

$$= a^b \log a - \frac{a}{b} \times a^b \quad (\text{since } g(b) = 0)$$

$$= a^b \left(\log a - \frac{a}{b} \right)$$



thus $\log a - \frac{a}{b} < 0$



hence $b < \frac{a}{\log a}$

Total for Question 21: 13 Marks

DNW